KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 18 · MATH

The Capital Street Survey

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Lines and Corners

Imani is helping her uncle, a land surveyor, measure street corners in Washington, D.C. Warm up before the lesson. Grab a protractor if you have one.

1. Review (Week 17): Divide. a) 8.4 ÷ 4 b) 52.6 ÷ 10

2. Draw a line, a line segment, and a ray. Label each one. (Hint: a line has arrows on both ends.)

3. Railroad tracks run side by side and never cross. The two rails are...

   parallel | perpendicular | intersecting at a square corner

4. An angle that is smaller than a square corner is called...

   acute | right | obtuse | straight

5. A full turn around a circle is 360 degrees. How many degrees is a quarter turn (1/4 of a full turn)?

6. Use a protractor, or the corner of a sheet of paper, to draw a right angle. Mark it with a small square in the corner.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Mapping the Corners

Show your work, and use a protractor where it helps.

1. Write acute, right, obtuse, or straight for each angle: a) 47° b) 90° c) 135° d) 180° e) 89°

2. Two paths start at the same point on a straight curb and split the straight angle into three angles. Two of them measure 38° and 75°. What does the third angle measure? Is it acute or obtuse?

3. Mr. Adeyemi's watch shows exactly 10:00. What is the smaller angle between the hour hand and the minute hand?

   30° 60° 90° 300°

4. Find the mistake: Tobin measured an obtuse angle and wrote 40°. In two or three complete sentences, explain what went wrong and what the angle really measures.

★ **Challenge:** A cyclist turns 1/4 of a full turn at one corner, then 1/12 of a full turn at the next corner, in the same direction. How many degrees did she turn in all? Is that total acute, right, or obtuse?

Looking Ahead: Week 19: you will visit a peach packing house in Fort Valley, Georgia, and find the volume of rectangular prisms with unit cubes, V = l × w × h, and V = B × h.